EXPERIMENT - 8

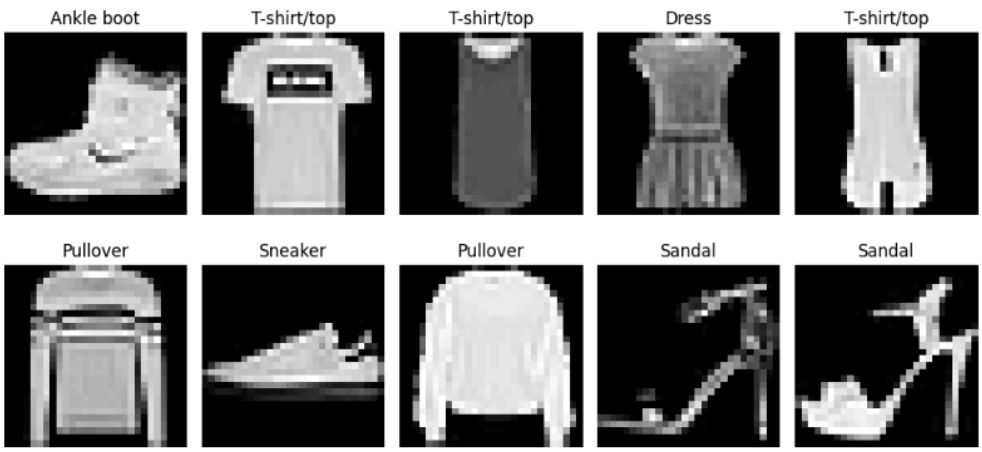
AIM: Design and train a Convolutional Neural Network (CNN) on image datasets (e.g., MNIST/CIFAR)

OUTPUT:

Load the dataset

```
Downloading data from https://storage.googleapis.com/tensorflow/tf-keras-datasets/train-labels-idx1-ubyte.gz
29515/29515 ━━━━━━━━━━━━━━━━━━━━ 0s 0us/step
Downloading data from https://storage.googleapis.com/tensorflow/tf-keras-datasets/train-images-idx3-ubyte.gz
26421880/26421880 ━━━━━━━━━━━━━━━━━━━━ 0s 0us/step
Downloading data from https://storage.googleapis.com/tensorflow/tf-keras-datasets/t10k-labels-idx1-ubyte.gz
5148/5148 ━━━━━━━━━━━━━━━━━━━━ 0s 0us/step
Downloading data from https://storage.googleapis.com/tensorflow/tf-keras-datasets/t10k-images-idx3-ubyte.gz
4422102/4422102 ━━━━━━━━━━━━━━━━━━━━ 0s 0us/step
Training Shape: (60000, 28, 28)
Testing Shape: (10000, 28, 28)
```

Visualize Sample Images



Build, compile and train CNN Model

```
Epoch 1/10
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 52s 27ms/step - accuracy: 0.8315 - loss: 0.4692 - val_accuracy: 0.8481 - val_loss: 0.4006
Epoch 2/10
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 50s 26ms/step - accuracy: 0.8841 - loss: 0.3160 - val_accuracy: 0.8837 - val_loss: 0.3262
Epoch 3/10
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 49s 26ms/step - accuracy: 0.9019 - loss: 0.2689 - val_accuracy: 0.8986 - val_loss: 0.2808
Epoch 4/10
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 82s 26ms/step - accuracy: 0.9111 - loss: 0.2387 - val_accuracy: 0.9011 - val_loss: 0.2778
Epoch 5/10
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 82s 26ms/step - accuracy: 0.9197 - loss: 0.2153 - val_accuracy: 0.9038 - val_loss: 0.2644
Epoch 6/10
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 80s 25ms/step - accuracy: 0.9282 - loss: 0.1943 - val_accuracy: 0.9024 - val_loss: 0.2751
Epoch 7/10
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 48s 26ms/step - accuracy: 0.9347 - loss: 0.1757 - val_accuracy: 0.9101 - val_loss: 0.2565
Epoch 8/10
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 49s 26ms/step - accuracy: 0.9396 - loss: 0.1604 - val_accuracy: 0.9024 - val_loss: 0.2686
Epoch 9/10
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 48s 26ms/step - accuracy: 0.9461 - loss: 0.1448 - val_accuracy: 0.9089 - val_loss: 0.2677
Epoch 10/10
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 49s 26ms/step - accuracy: 0.9502 - loss: 0.1324 - val_accuracy: 0.9133 - val_loss: 0.2620
```

Evaluate Model

```
313/313 ━━━━━━━━━━━━━━━━━━━━ 2s 7ms/step - accuracy: 0.9133 - loss: 0.2620
Test Accuracy: 0.9132999777793884
```

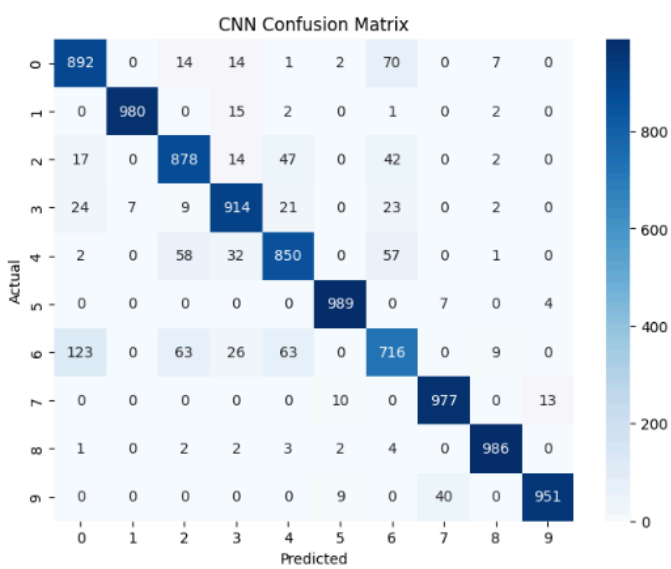
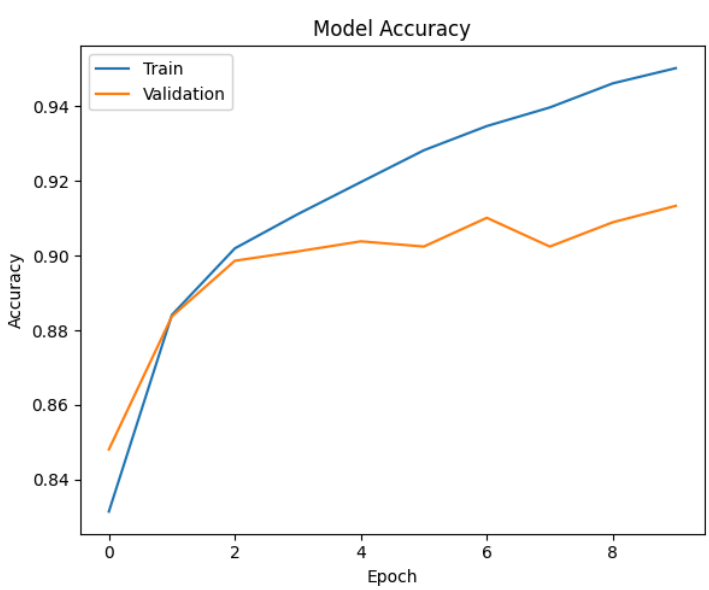
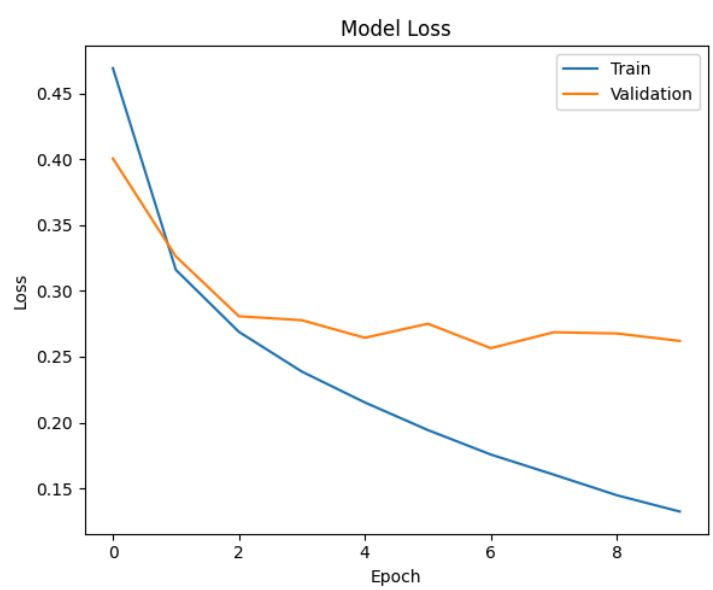
Predictions

```
313/313 ━━━━━━━━━━━━━━━━━━━━ 2s 7ms/step
              precision    recall  f1-score   support

           0       0.84      0.89      0.87      1000
           1       0.99      0.98      0.99      1000
           2       0.86      0.88      0.87      1000
           3       0.90      0.91      0.91      1000
           4       0.86      0.85      0.86      1000
           5       0.98      0.99      0.98      1000
           6       0.78      0.72      0.75      1000
           7       0.95      0.98      0.97      1000
           8       0.98      0.99      0.98      1000
           9       0.98      0.95      0.97      1000

    accuracy                           0.91     10000
   macro avg       0.91      0.91      0.91     10000
weighted avg       0.91      0.91      0.91     10000
```

Confusion matrix



PLot accuracy and loss